

$$= \frac{1}{9} \text{ of } \frac{1}{11} \text{ of } \frac{1}{14} \text{ of } 693$$

$$= \frac{1}{9} \text{ of } \frac{1}{11} \text{ of } 49.5$$

$$= \frac{1}{9} \text{ of } 4.5 = 0.5$$

36. (d)

$$\frac{2.25^2 - 1.25^2}{2.25 - 1.25} = \frac{(2.25 - 1.25)(2.25 + 1.25)}{(2.25 - 1.25)}$$

$$= (2.25 + 1.25) = 3.5$$

37. (a) Let the speed of the train is V

Distance covered = (relative speed of the train) $\times$ time

$$\Rightarrow (100 + 120) = (50 + V) \times 6$$

$$\Rightarrow \frac{220}{1000} \text{ km} = (5 + V) \times \left(\frac{3}{3600} \right) \text{ hr}$$

$$\Rightarrow 22 = (50 + V) \times \frac{1}{6}$$

$$\Rightarrow V = 132 - 50 = 82 \text{ kmph}$$

38. (c) Time taken by train 1 = 4 hours

$$\text{Time taken by train 2} = 3.5 = \frac{7}{2} \text{ hours}$$

Let the distance between Meerut and Delhi

$$\text{is } 28 \text{ km (LCM of } 4 \text{ \& } \frac{7}{2} \text{)}$$

$$\text{Speed of train 1} = \frac{28}{4} = 7 \text{ km/hr}$$

$$\text{Speed of train 2} = \frac{28}{3.5} = 8 \text{ km/hr}$$

Relative speed = Speed of train 1 + Speed of train 2

$$\Rightarrow \text{Relative speed} = 7 + 8 = 15 \text{ km}$$

Distance travel by train 1 before starting off train 2 = speed of train 2 $\times$ time

$$\Rightarrow 7 \times 2 = 14 \text{ km}$$

$$\text{Remaining distance} = 28 - 14 = 14 \text{ km}$$

Time to cross each other

$$\frac{\text{remaining distance}}{\text{relative speed}} = \frac{14}{15} \text{ or } 56 \text{ minutes}$$

$$\text{Total time is } = 2 \text{ hour} + 56 \text{ min}$$

39. (b) According to question,

Simple interest is the same in both conditions

$$\Rightarrow P \times 8 \times \frac{5}{100} = P \times 5 \times \frac{R}{100}$$

$$\Rightarrow \frac{40}{100} = \frac{R}{20}$$

$$\Rightarrow R = 8\%$$

40. (a) Total age of man and his son

$$= 40 \times 2 = 80 \text{ years}$$

Let the age of man and son be $11x$ and $5x$.

$$\Rightarrow 11x + 5x = 80$$

$$\Rightarrow x = 5 \text{ years}$$

$$\text{Age of son} = 5x$$

$$\Rightarrow 5 \times 5 = 25 \text{ years}$$

41. (c) Converting the fractions into decimal

$$\Rightarrow \frac{3}{4} = 0.75$$

$$\Rightarrow \frac{10}{13} = 0.77$$

$$\Rightarrow \frac{11}{14} = 0.79$$

$$\Rightarrow \frac{7}{9} = 0.78$$

42. (a) Sum of last 3 numbers = $3 \times 15 = 45$

$$\text{Sum of 2 middle number} = 45 - 21 = 24$$

$$\text{Sum of first 3 number} = 3 \times 16 = 48$$

$$\therefore \text{First number} = 48 - 24 = 24$$

43. (a) The cost price of 2 dozen bananas

$$= 2 \times 16 = ₹ 32$$

$$\text{SP of 18 bananas} = \frac{18}{12} \times 12 = ₹ 18$$

$$\text{SP of remaining bananas} = \frac{6}{12} \times 4 = ₹ 2$$

$$\text{Total SP} = 18 + 2 = ₹ 20$$

$$\text{Loss \%} = \frac{32 - 20}{32} \times 100 = 37.5\%$$

44. (a) Quantity of pure liquid in the final

$$\text{mixture} = A \left[1 - \left(\frac{B}{A} \right) \right]^n$$

Where A is Quantity of pure milk

B is quantity of milk replaced by water

n represent how many times water replaced by milk = 3

Putting the value of A and B in above equation

$$= 40 \times \left[1 - \left(\frac{4}{40} \right) \right]^3$$

$$= 40 \times \left[1 - \left(\frac{1}{10} \right) \right]^3$$

$$= 40 \times \left(\frac{729}{1000} \right) = 29.16 \text{ litres}$$

45. (d) Number of possible outcomes

$$= 6^2 = 36$$

Sum of number which is divisible by 4 or 6.

$$\{(1, 3), (1, 5), (2, 2), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (5, 1), (5, 3), (6, 2), (6, 6)\}$$

Number of favorable outcomes = 14

$$\text{Probability} = \frac{14}{36} = \frac{7}{18}$$

46. (d) Let the required fraction is $\frac{x}{y}$

According to the question,

$$y = 2x + 4$$

...(i)

Again, According to the question

$$(y - 6) = 12(x - 6)$$

$$\Rightarrow y - 6 = 12x - 72$$

From equation (i),

$$(2x + 4) - 6 = 12x - 72$$

$$\Rightarrow 10x = 70$$

$$\Rightarrow x = 7$$

Again, From the equation (i),

$$y = 2 \times 7 + 4$$

$$\Rightarrow y = 18$$

So, the fraction will be $\frac{7}{18}$.

47. (d) Let the first and second parts be x and y respectively.

According to the question,

$$\Rightarrow \left(\frac{x}{5} \right) : \left(\frac{y}{8} \right) = \frac{3}{4}$$

Cross Multiplying the above equation

$$\Rightarrow \frac{8x}{5y} = \frac{3}{4}$$

$$\Rightarrow \frac{x}{y} = \frac{15}{32} = z \quad \dots\dots(z \text{ is some constant})$$

$$\Rightarrow x = 15z, y = 32z$$

Sum of two parts = 94

$$x + y = 94$$

$$\Rightarrow 15z + 32z = 94$$

$$\Rightarrow 47z = 94 \text{ or } z = 2$$

$$\text{Hence, } x = 15z = 15 \times 2 = 30$$

48. (c) The area of the field = πr^2

$$\Rightarrow \pi(r)^2 = 13.86$$

$$\Rightarrow \left(\frac{22}{7} \right) r^2 = 13.86 \times 10000$$

$$[\because 1 \text{ hectares} = 10000 \text{ m}^2]$$

$$\Rightarrow r^2 = \frac{7}{22} \times 138600$$

$$\Rightarrow r^2 = 44100$$

$$\Rightarrow r = 210 \text{ m}$$

Length of the circular field = $2\pi r$

$$= 2 \times \frac{22}{7} \times 210 = 1320 \text{ m}$$

Cost of fencing the circular field

$$= 1320 \times 4.4 = 5808$$

49. (d) Efficiency of P = $\frac{1}{8}$

$$\text{Efficiency of Q} = \frac{1}{10}$$

$$\text{Efficiency of R} = \frac{1}{12}$$

Work done by P + Q + R in 1 hour

$$= \frac{1}{8} + \frac{1}{10} + \frac{1}{12} = \frac{(15 + 12 + 10)}{120}$$

$$= \frac{37}{120}$$

P, Q, and R work for 2 hours 9 A.M to

$$11 \text{ A.M} = \frac{37}{120} \times 2 = \frac{37}{60}$$

$$\text{Remaining work is } = 1 - \frac{37}{60} = \frac{23}{60}$$

Work done by Q and R in an hour

$$= \left(\frac{23}{60} \right) \div \left(\frac{11}{60} \right)$$

$$= \left(\frac{23}{60} \right) \times \left(\frac{60}{11} \right) = \frac{23}{11} \approx 2 \text{ hour}$$

So, add 2 hours in 11 A.M we get 1 P.M

50. (b) The angle between the hour-hand and minute-hand

$$\theta = \left(30H - \frac{11M}{2} \right)$$

When minute and hour hands coincide then,

$$\theta = 0$$

$$\Rightarrow 0 = 30 \times 3 - \frac{11M}{2}$$

$$\Rightarrow 0 = 90 - \frac{11M}{2}$$

$$\Rightarrow 90 = \frac{11M}{2}$$

$$\Rightarrow M = 16 \frac{4}{11} \text{ minutes}$$

So, the both hand will coincide at

$$3 \text{ hour } 16 \frac{4}{11} \text{ min}$$

51. (c) The Shanti Swarup Bhatnagar Prize for **Science and Technology** is one of the most prestigious scientific awards in India.

The award was established in 1958 in honor of Dr. Shanti Swarup Bhatnagar, a renowned Indian scientist, educationist, and the founder-director of the Council of Scientific and Industrial Research (CSIR).

The primary objective of the Shanti Swarup Bhatnagar Awards is to recognize outstanding contributions to various fields of science and technology.

52. (d) **INS Chakra** refers to the nuclear-powered submarine operated by the Indian Navy.

It's worth noting that India has leased nuclear-powered submarines from Russia under the name INS Chakra.

The INS Chakra is part of a series of submarines that India has acquired from Russia over the years.

53. (b) The DRDO (Defence Research and Development Organization) developed anti-tank missile capable of being launched from a helicopter is the **Helina (HELicopter-launched Nag)**.

Helina is a third-generation, fire-and-forget, anti-tank guided missile designed to be launched from helicopters.

Helina is equipped with an infrared imaging seeker that allows it to lock onto and track targets independently, eliminating the need for continuous guidance from the launch platform.

54. (c) **Covaxin** is an indigenously developed COVID-19 vaccine produced by Bharat Biotech, an Indian biotechnology company.

It was developed in collaboration with the Indian Council of Medical Research (ICMR) and the National Institute of Virology (NIV).

Covaxin is an inactivated virus vaccine, and it received emergency use authorization in India.

Pfizer is a multinational pharmaceutical company, and the Pfizer-BioNTech COVID-19 vaccine is not indigenously developed in India.

55. (a) Asad Ali Khan (1937–2011) was a renowned Indian musician, and he was a master of the **Rudra veena**, a traditional plucked string instrument associated with Indian classical music.

He hailed from the Senia gharana, a school or tradition of music that has been associated with the dhrupad style, one of the oldest forms of classical music in North India.

Asad Ali Khan was born in a family with a rich musical heritage. He received training in music from his father and grandfather, both of whom were accomplished musicians.

56. (d) India and the **United States** have signed a Project Agreement for Air-Launched Unmanned Aerial Vehicle (UAV).

The agreement, signed between the Ministry of Defence (India) and the US Department of Defence, falls under the framework of the Defence Technology and Trade Initiative (DTTI).

57. (c) “Bahishkrit Bharat” was a Marathi language newspaper launched by **Dr. B. R. Ambedkar** in 1927.

Dr. Ambedkar, a prominent social reformer, jurist, and the chief architect of the Indian Constitution, used this newspaper as a platform to address the social issues and concerns of the marginalized and oppressed communities in India.

The term “Bahishkrit” translates to “Excluded” or “Outcast,” and the newspaper played a crucial role in advocating for the rights and upliftment of the Dalits and other socially disadvantaged groups.

58. (a) Raja Krishnadevaraya, who ruled the Vijayanagara Empire in the 16th century, was a patron of literature and arts.

His reign is often referred to as the “Golden Age of Telugu Literature.”

During this period, there was significant literary and cultural development in the

Telugu language, and many poets and scholars flourished under his patronage.

59. (d) “Why I am an Atheist” is an essay written by Bhagat Singh, the Indian revolutionary and freedom fighter.

The essay reflects Bhagat Singh's thought on religion, atheism, and his reasons for rejecting religious beliefs.

He wrote this essay in 1930 while he was in Lahore Central Jail, awaiting trial for his involvement in the killing of British police officer J.P. Saunders.

60. (d) A supernova is a powerful and catastrophic explosion that occurs in the final stages of a massive star's life cycle.

This event results in a sudden and intense increase in brightness, outshining an entire galaxy for a brief period.

The energy released during a supernova is immense, and it can briefly outshine an entire galaxy.

61. (c) The Attorney General of India is the principal legal advisor to the Government of India and is appointed by the President of India.

The Attorney General has the right to represent the government in legal matters and has the authority to appear in any court of law, including the Supreme Court.

The Attorney General of India is the highest law officer in the country. This position is established under Article 76 of the Indian Constitution, and the Attorney General is appointed by the President of India.

62. (a) **Sumit Antil** won the Gold medal in Javelin Throw at the 2020 Paralympics.

He is an Indian para-athlete who set a new world record in the Men's Javelin Throw F64 category during the Paralympics held in Tokyo in 2021.

Sumit's impressive performance and world record made him a gold medalist in the event.

63. (b) “Global competitiveness Report” is released by WEF. WEF stands for the World Economic Forum.

The **World Economic Forum (WEF)** is an international organization that brings together leaders from various sectors, including business, government, academia, and civil society.

The WEF is best known for its annual meeting in Davos, Switzerland, where leaders gather to discuss and deliberate on pressing global issues.

64. (d) Semiconductors are the materials which have conductivity between insulators and conductors.

They can be compounds or pure elements.

Out of the given options, **Si** is one of the most commonly used semiconductor.